

Relativistic Quantum Field Theory I(PH.60053)

Exercise Set 1

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Note: I generally disdain the use of natural units in undergraduate studies. For students having gone through a symbolic mess of a calculation, something as simple as dimensional analysis gives a good sanity check. And while this course is explicitly designed for graduate students, I would like to stand by this conviction and perform all calculations with explicit engineering dimensions and constants. I believe it is reasonably simple for the grader to convert it into natural units. I appreciate your understanding.

Problem 1

We use the following definitions:

$$A^\mu = \left(\frac{\phi}{c}, \vec{A}\right), \quad F^{\mu\nu} = \partial^{[\mu} A^{\nu]}, \quad J^\mu = (-c\rho, \vec{J}),$$

$$E^i = -\partial_i \phi - \partial_t A^i = c \left(\left(-\frac{1}{c} \partial_t\right) A^i - \partial_i \left(\frac{\phi}{c}\right) \right) = cF^{0i},$$

$$B^i = \partial_j A^k - \partial_k A^j = F^{jk} \quad (\epsilon_{ijk} = +1).$$

We start with the Lagrangian density:

$$\mathcal{L} = -\frac{1}{4\mu_0} F_{\alpha\beta} F^{\alpha\beta} + J_\alpha A^\alpha.$$

We find the following partial derivative relations:

$$\frac{\partial(\partial_\alpha A_\beta)}{\partial(\partial_\mu A_\nu)} = \delta_\alpha^\mu \delta_\beta^\nu \Rightarrow \frac{\partial(\partial^\alpha A^\beta)}{\partial(\partial_\mu A_\nu)} = \eta^{\mu\alpha} \eta^{\nu\beta}$$

$$\Rightarrow \frac{\partial}{\partial(\partial_\mu A_\nu)} (F_{\alpha\beta} F^{\alpha\beta}) = (\delta_\alpha^\mu \delta_\beta^\nu - \delta_\beta^\mu \delta_\alpha^\nu) F^{\alpha\beta} + F_{\alpha\beta} (\eta^{\mu\alpha} \eta^{\nu\beta} - \eta^{\mu\beta} \eta^{\nu\alpha}) = 4F^{\mu\nu}$$

Thus, the Euler-Lagrange equations are given by

$$0 = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} \right) - \frac{\partial \mathcal{L}}{\partial A_\nu} = \partial_\mu \left(-\frac{1}{\mu_0} F^{\mu\nu} \right) - J^\nu$$

$$\therefore \partial_\mu F^{\mu\nu} = \mu_0 J^\nu$$

$$1. \quad \nu = 0: \quad \partial_\mu F^{\mu 0} = -\frac{1}{c} \vec{\nabla} \cdot \vec{E} \Rightarrow \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$2. \nu = 1: \partial_\mu F^{\mu 1} = \left(\frac{1}{c}\partial_t\right)\left(\frac{E_x}{c}\right) + \partial_y(-B^z) + \partial_z B^y = \left(\frac{1}{c^2}\partial_t \vec{E} - \vec{\nabla} \times \vec{B}\right)^x.$$

Thus, generally, $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \frac{1}{c^2} \partial_t \vec{E}$.

Furthermore, the Bianchi identity can be expressed as follows:

$$1. \partial_{[0} F_{12]} = \frac{1}{c} \partial_t B^z + \partial_x \left(+\frac{E_y}{c}\right) + \partial_y \left(-\frac{E_x}{c}\right) = \frac{1}{c} \left(\partial_t \vec{B} + \vec{\nabla} \times \vec{E}\right)^z.$$

Thus, generally, $\vec{\nabla} \times \vec{E} = -\partial_t \vec{B}$.

$$2. \partial_{[1} F_{23]} = \partial_x B^x + \partial_y B^y + \partial_z B^z = \vec{\nabla} \cdot \vec{B} = 0.$$

Thus, we see that the Euler-Lagrange equations as well as the Bianchi identity successfully reproduce the conventional Maxwell equations. $\square$

Problem 2

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